

e. Discussion:

i. Based on the weights of the linear regression model, rank the features (note: only continuous features) in decreasing order of influence on the predictions.

Printing the feature importance:

	Feature	Importance
7	median_income	0.647880
1	latitude	0.464731
0	longitude	0.457954
4	total_bedrooms	0.384863
5	population	0.350110
6	households	0.127707
3	total_rooms	0.122626
2	housing_median_age	0.120422
9	ocean_proximity_INLAND	0.108360
8	ocean_proximity_<1H OCEAN	0.055817
12	ocean_proximity_NEAR OCEAN	0.047156
10	ocean_proximity_ISLAND	0.021126
11	ocean_proximity_NEAR BAY	0.020495

ii. Discuss how the influence of the features (obtained in question e.1) relates to the pair-wise correlation results calculated above i.e., are the features that are highly correlated with the output also the most influential or is there some other phenomenon being observed?

In isolation the most correlated features is often the most influential in a linear regression model however when combined with other features the combined affect may change the influence of the single feature.

iii. Comment about the MAPE, RMSE, R2 results. What can we learn from each of these results about the model prediction performance?

- MAPE gives the average error as a percentage of true values. A lower MAPE means better relative accuracy.
- RMSE gives the typical size of the error with large errors penalized more strongly due to squaring. Lower RMSE can mean that the predictions tend to lie closer to the true values.

- R^2 measures the proportion of the variance in the target that the model explains relative to the baseline. A value close to 1 indicates that the model is capturing most of the variability.

iv. Why is centering and scaling the data important before performing PCA?

The reason centering and scaling are important is because PCA is driven by the covariance matrix which is sensitive to means and variable scales. Centering helps to avoid the principal component being dominated by the overall mean rather than meaningful variation. Scaling is important because if data is in different scales, more meaning will be given to larger values.